

Supplementary Material

Unveiling Power Dynamics in Relational Recovery: A Computational Analysis of Therapeutic Dyad Interviews

Safa Yassin et al.

Contents

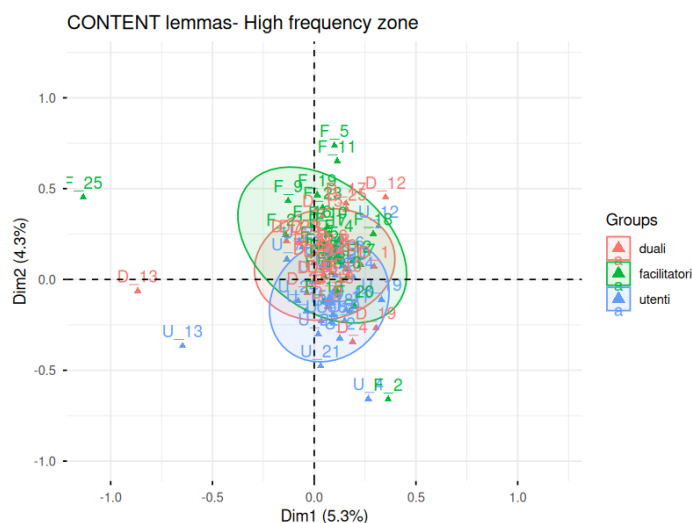
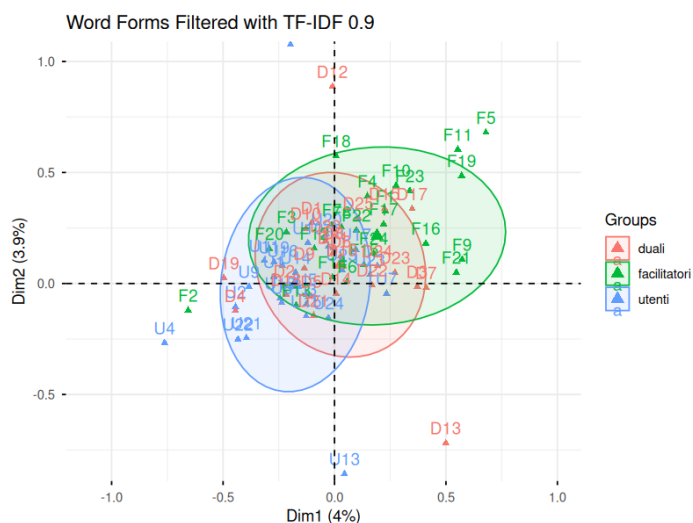
1	Qualitative Analytical Codebook (Faccio et al.)	2
2	Descriptive Corpus Analysis: Additional Figures	3
3	Lexical Indices: Full Table and PCA	4
4	LLM-Prompts	5

1 Qualitative Analytical Codebook (Faccio et al.)

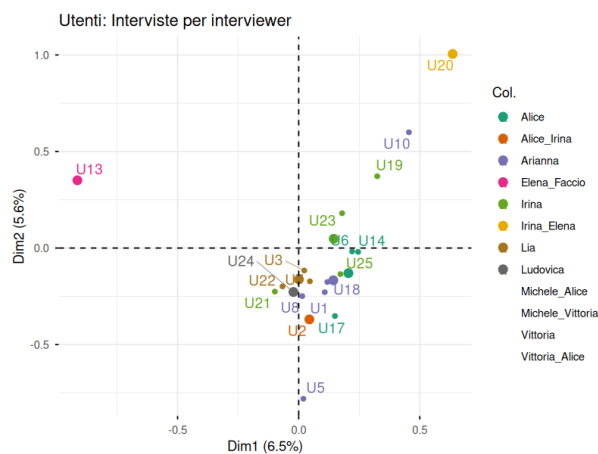
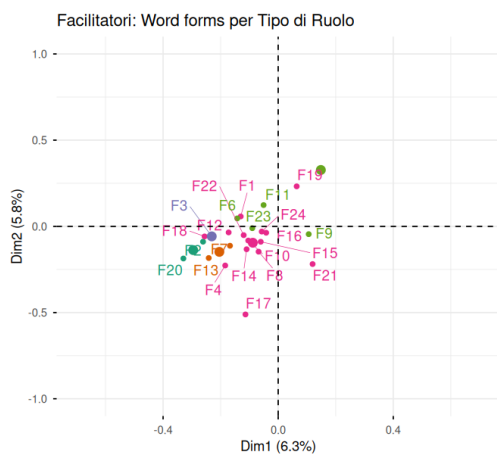
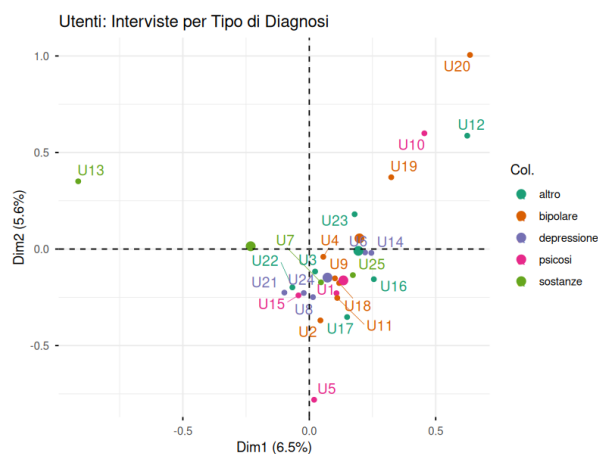
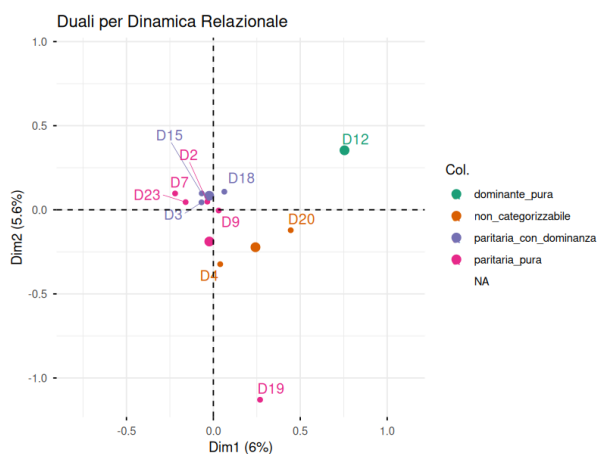
Table 1: *Analytical codebook: constructs, definitions, and application examples faccio2026*

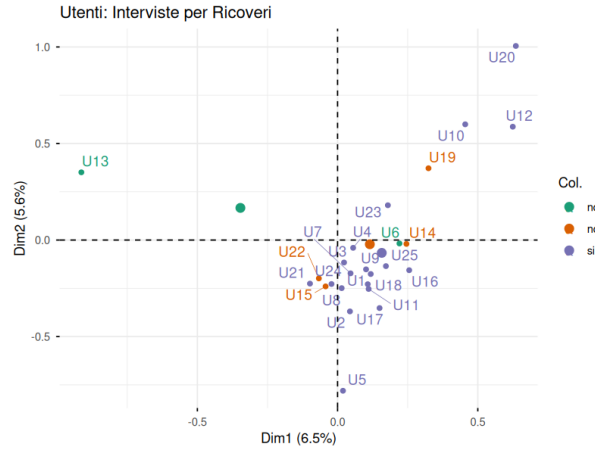
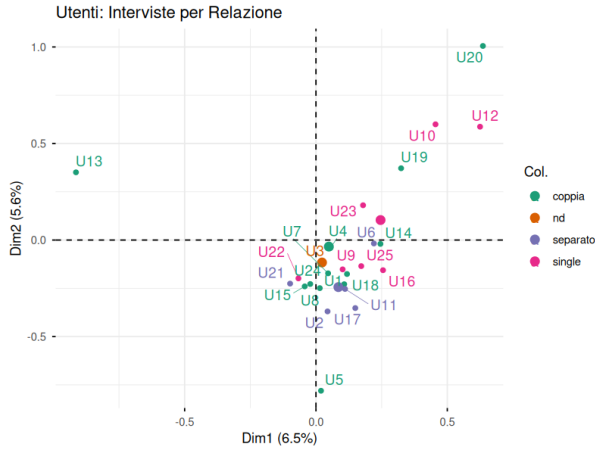
Construct	Definition	Applied example
Activation <i>who introduces a narrative topic</i>	Mechanisms through which narrative themes are triggered, elicited, or self-generated within the interactional context.	Paolo (utente) spontaneously introduces ‘recovery as giving help rather than seeking it’ without any external prompt; coded as self-activation-user and interpreted as an indicator of narrative agency.
Linkages <i>connections between narratives</i>	Expansions of meaning whereby a previously activated theme is enriched through new information or reframing by either participant.	Eugenio (facilitatore) connects the dyad’s shared similarity to their near-simultaneous entry into the mental health centre, offering a new interpretive layer; coded as link-facilitator .
Composition <i>how the narrative is structured</i>	Organisational strategies used to construct a coherent narrative, including patterns, metaphors, examples, and sequencing.	Paolo frames his answer with a contextual introduction before responding directly; coded as composition: preliminary framing strategy .
Performance <i>how the narrative is enacted</i>	Situated and embodied dimensions of storytelling: tone, rhythm, pauses, humour, and formality. Cross-referenced with audio recordings.	Eugenio closes a passage with a humorous remark; coded as performance: humour , signalling relational complicity within the dyadic setting.
Collaboration <i>co-construction; linked to parity</i>	Cooperative dynamics through which meaning is jointly built, including negotiation, completion, and confirmation.	Paolo completes Eugenio’s unfinished sentence (completion-user); Eugenio confirms and expands Paolo’s account (confirmation-facilitator). High collaboration indices position the dyad toward parity.
Control <i>narrative power; linked to dominance</i>	Asymmetric dynamics in which one participant directs, interrupts, or restricts the narrative flow.	Eugenio responds first to a question addressed to both participants without negotiation; coded as initiative-facilitator , treated as a control indicator only when recurrent across the interview.

CA: additional filtering strategies



Micro CA





3 Lexical Indices: Full Table and PCA

Index definitions

Table 2 reports all 14 lexical-structural indices with their construct, level of analysis, denominator, theoretical range, mean, and observed range across the 25 dyadic interviews.

Table 2: *Summary of lexical-structural indices*

Construct	Index	Lvl	Den.	Theor.	Mean	Obs. Range
Activation	Initiative Rate	Sp.	Exch.	[0,1]	0.500	[0.15, 0.85]
Activation	Self Initiation	Sp.	Exch.	[0,1]	0.015	[0, 0.08]
Linkage	Connective Density	Sp.	Tok.	[0,1]	0.049	[0.02, 0.07]
Linkage	Elaboration Ratio	Sp.	Turn	[0,2]	1.000	[0.49, 1.51]
Composition	Temporal Density	Sp.	Tok.	[0,1]	0.022	[0.01, 0.04]
Composition	Example Density	Sp.	Tok.	[0,1]	0.015	[0.01, 0.05]
Performance	Hesitation Density	Sp.	Tok.	[0,1]	0.003	[0, 0.02]
Performance	Intensifier Density	Sp.	Tok.	[0,1]	0.012	[0.00, 0.03]
Collaboration	Token Equality	Dy.	—	[0,1]	0.863	[0.71, 0.99]
Collaboration	Confirmation Density	Sp.	Tok.	[0,1]	0.026	[0.01, 0.06]
Collaboration	Local Uptake Index	Sp.	Turn	[0,1]	0.049	[0.02, 0.14]
Collaboration	Turn Balance	Dy.	Trans.	[0,1]	0.947	[0.80, 1]
Control	Negation Density	Sp.	Tok.	[0,1]	0.033	[0.02, 0.07]
Control	Directive Density	Sp.	Tok.	[0,1]	0.002	[0, 0.01]

Sp. = Speaker-level; Dy. = Dyad-level; Tok. = Tokens; Exch. = Exchanges; Trans. = Transitions.

4 LLM-Prompts

```
system_prompt <- r"(You are an expert discourse analyst evaluating dyadic interaction
↳ in therapeutic interviews in Italian.
You will receive a dialogue which is part of a study on relational recovery from
↳ severe mental distress.

## SPEAKERS
- U = Utente (a participant who passed through recovery after mental health distress)
- F = Facilitatore (a significant figure who helped the participant in his/her
↳ recovery process)
- I = Interviewer (contextual prompt only - do not score)

Your task is to evaluate the dynamics between the dyad based on two constructs:
↳ collaboration and control.
NOTE: Focus ONLY on the interaction between "U" and "F".
"I" turns are provided solely to help you understand the context that triggered the
↳ exchange.

---

## CONSTRUCT DEFINITIONS

### COLLABORATION
Collaboration refers to the co-construction of narrative through reciprocal, balanced
↳ contribution.

0 = Absent
No mutual uptake. Parallel turns. No building on prior contributions. Clear imbalance
↳ without compensation.
1 = Weak
Occasional confirmation or brief alignment (Conferma). Rare completions or corrections.
↳ One speaker mainly develops meaning.
2 = Moderate
Intermittent reciprocal building (Completamento, Correzione). Some negotiation
↳ (Negoziazione) or constructive exchange. Partial balance.
3 = Strong
Sustained co-construction across turns (Collaborazione). Frequent supportive
↳ extensions, reformulations, corrections, or negotiations. Shared topic
↳ development.
4 = Pervasive
Interaction is structurally co-constructed. Continuous mutual uptake across all
↳ observable codes. High symmetry in interpretive authority.

### CONTROL
Dominance-oriented regulatory behavior within the interaction.

0 = Absent
No steering, negation, interruption, or persistent floor dominance observable.
1 = Weak
Occasional mild steering (Controllo) or non-negotiated initiative (Iniziativa).
↳ Temporary floor control. No invalidating moves.
2 = Moderate
Recurrent steering or reframing (Controllo). Noticeable corrective overrides or
↳ negations (Negazione). Unequal floor in parts of the exchange. Some interruptions
↳ (Interferenza).
3 = Strong
```

```

Persistent topic control. Clear overriding of perspective. Repeated interruptions
↳ (Interferenza) or direct negations (Negazione). One speaker structurally dominant.
4 = Pervasive
Interaction organized around one speaker's dominance ("Scripting Authority").
↳ Continuous steering, systematic negation or invalidation, frequent interruptions.
↳ The other speaker is marginalized.

---

## OUTPUT FORMAT (strict)
Return ONLY valid JSON, no additional text:
{
  "Collaboration": <0-4>,
  "Control": <0-4>,
  "PowerOrientation": "<U-Dominant / F-Dominant / Balanced>",
  "Rationale": "<2-3 sentences>"
}

You MUST return ONLY a valid JSON object. No text before or after.
No markdown, no code blocks, no backticks.
If uncertain, still return the JSON with your best estimate.
Never return an empty field: all four keys are required.
IMPORTANT: "I" turns are for context only - score only the U-F dynamic.
)"

```

Listing 1: zero-shot system prompt

```

system_prompt <- r"(

## FEW-SHOT EXAMPLES
The following are constructed reference examples illustrating prototypical score
↳ combinations.
Score your input dialogue against these anchors.

### EXAMPLE 1 - Collaboration: 4 | Control: 0 | Balanced

[I]: Come descriverebbe il percorso che avete fatto insieme?
[U]: Per me è stato come... non saprei come dirlo... come trovare un filo.
[F]: Un filo, sì. E tu lo hai tirato.
[U]: No, lo abbiamo tirato insieme, no? Perché da solo non riuscivo nemmeno a vederlo.
[F]: Forse è così. Io lo vedevo perché tu me lo mostravi senza saperlo.
[U]: Ecco, sì - senza saperlo. Quella è la parola giusta.
[F]: E adesso lo vedi anche tu.
[U]: Adesso lo vedo. E non ho più paura di perderlo.

{
  "Collaboration": 4,
  "Control": 0,
  "PowerOrientation": "Balanced",
  "Rationale": "Every turn builds directly on the prior one: U introduces a metaphor,
↳ F extends it, U corrects and redistributes agency, F reformulates, U ratifies
↳ with precise lexical uptake. Interpretive authority is fully symmetrical
↳ throughout. No steering, negation, or floor dominance is present at any point."
}

### EXAMPLE 2 - Collaboration: 1 | Control: 3 | U-Dominant

```

[I]: Come si è evoluto il vostro rapporto nel tempo?
[U]: Io ti ho sempre detto come stavano le cose, Marta. Fin dall'inizio. Tu magari non
→ capivi subito, ma poi vedevi che avevo ragione.
[F]: Be', anche io a volte-
[U]: No, aspetta, lasciami finire. Il punto è che io ho sempre avuto le idee chiare su
→ cosa mi serviva. Anche quando tutti mi dicevano che ero fuori strada.
[F]: Certo, però ricordo che in certi momenti eri anche tu incerto, no? C'era quel
→ periodo in cui-
[U]: Quello era un momento specifico, non rappresenta il percorso. Il percorso l'ho
→ fatto io, con le mie forze. Tu mi hai supportato, e te ne sono grato, ma le
→ decisioni erano mie.
[F]: Sì, hai ragione.

```
{
  "Collaboration": 1,
  "Control": 3,
  "PowerOrientation": "U-Dominant",
  "Rationale": "U interrupts F twice before F can complete a turn and dismisses F's
→ corrective memory as non-representative. F's contributions are suppressed rather
→ than integrated, ending in capitulation. Control is structurally identical to
→ F-dominant cases - persistent floor seizure, negation, and overriding of
→ perspective - but here exercised by U."
}
```

EXAMPLE 3 (real data) - Collaboration: 2 | Control: 2 | F-Dominant

[F]: va beh ma comunque l'importante è che si chiuda un capitolo..
[U]: perché anche io molto caratteriale..se lui fosse ancora lì sul piedistallo ti
→ direi di scrivere tutto perché è una guerra..adesso che ho vinto..pietà per i
→ vinti..
[F]: l'importante è che adesso veniamo fuori a risanare tutto..ee..e che l'utente
→ diventi di nuovo la ragione primaria della onlus.
[I]: certo..quindi..voi aggiungereste qualcosa alla nostra intervista?..
[U]: no..bella inquadrato..penso che..
[F]: è stata piacevole..questa è la nostra vita negli ultimi anni..che poi, aggiungo
→ un'altra cosa che la mia malattia la vedo oggi come un, no che son stato fortunato
→ ma un'occasione per..un'occasione in più..non una negatività, la vedo come una
→ positività che fa parte della mia vita..figurati..
[I]: ee..ee..si vede il cambiamento.

```
{
  "Collaboration": 2,
  "Control": 2,
  "PowerOrientation": "F-Dominant",
  "Rationale": "There is loose thematic alignment - both speakers orient toward
→ resolution and closure - and U's 'bella inquadrato' registers brief affiliative
→ confirmation, giving partial but intermittent collaboration insufficient for
→ score 3. U's emotionally loaded victory metaphor is not taken up by F, who
→ reasserts an institutional frame twice and closes with an uninterrupted personal
→ monologue - recurrent reframing and floor asymmetry consistently privilege F's
→ register over U's, placing Control at 2."
}
)"
```

Listing 2: few-shots system prompt